

TriAgent: Divergence-Aware Multi-Agent Committees for Cost-Efficient and Privacy-Preserving Financial Sentiment Analysis

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Abstract

Production deployments of LLM-based financial sentiment analysis face a structural cost problem: most queries are trivially classifiable yet are routinely processed by expensive cloud reasoners, and the bill scales linearly with user count. We present **TriAgent**, a divergence-aware multi-agent committee that stacks three increasingly contextual tiers — a word-level lexicon agent (VADER), a sentence-level domain transformer (FinBERT), and a cross-sentence reasoner (Qwen2.5-Instruct, swept from 0.5 B to 14 B-4bit and across families) — and routes queries by a three-way *Semantic Divergence Index* (SDI). On Financial PhraseBank we find: (i) the domain specialist alone ($F1=0.88$) outperforms every generic LLM up to 7 B ($F1=0.81$) at $60\times$ lower cost; (ii) when the LLM is re-tasked as a *critic* over the smaller agents’ outputs, $F1$ plateaus at ≈ 0.87 across model sizes from 1.5 B to 7 B — “*interaction substitutes for parameters*” — but this plateau is granularity-stratified, not multi-agent-per-se: a 3-persona Qwen-1.5B vote drops to $F1=0.66$; (iii) a Shared Consensus Dictionary (SCD) using multilingual sentence-BERT serves as a cross-lingual canonicalizer, answering 95% of Chinese queries from an English cache at $F1=0.99$; (iv) the same SDI signal that gates routing doubles as a post-hoc LLM-hallucination detector at $AUC=0.90$. A 20-ticker backtest shows the SDI two-stage strategy attains the best risk-adjusted return (Sharpe=3.50) among all evaluated strategies, and a scale analysis shows TriAgent saves \$9.3 M/year vs. the default-to-cloud-LLM baseline at 10 M users. The framework is fully open-source and edge-deployable on a single 24 GB GPU.

1 Introduction

A mid-sized asset manager with 1 M users running 10 sentiment queries each per trading day generates ~ 3.6 B inference calls per year. At the public price of GPT-4o-mini that single workload costs $\sim \$1.1$ M/year; at GPT-4 prices, $\sim \$36.5$ M/year (Figure 1). The cost scales linearly with user

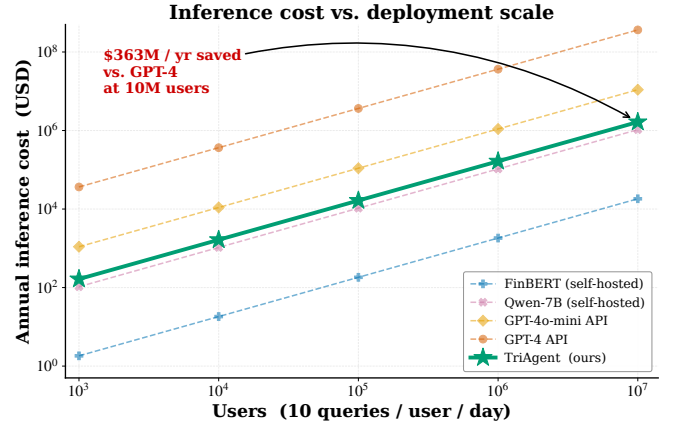


Figure 1: Annual inference cost vs. user count (10 queries/user/day). At 10 M users, default-to-cloud-LLM costs \$11 M–\$365 M/yr; TriAgent shaves this to \$1.65 M/yr, a \$9.3 M/yr saving against GPT-4o-mini.

count while the value of each LLM call falls — most domain queries are trivially classifiable. **Token efficiency** — spending fewer tokens per query, on cheaper agents, with fewer escalations — is therefore the first-order operational concern, not an afterthought. The CFP for FinLLM@IJCAI 2026 calls out exactly this regime: multi-agent cloud-edge collaboration, token economics, and cross-border fairness in deployment.

Running example. Throughout we trace one FPB sentence: “... its net profit halved to 1.2 mln euro... from 2.2 mln euro...” (gold = negative). VADER (+0.44) and FinBERT (+0.53) are both fooled by the surface word *profit*; Qwen-7B alone correctly outputs negative (−0.80, “profit declined”); critic@1.5B and critic@7B are *anchored* on V+F’s positives and stay wrong; only round-2 debate, which feeds back the LLM’s own R1 rationale, recovers negative. This single example threads through Sections 3–6.

Contributions. TRIAGENT is a divergence-aware committee whose three tiers correspond to three contextual granularities (word → sentence → cross-sentence). We contribute: (C1) a three-way Semantic Divergence Index (SDI) with a four-quadrant routing interpretation; (C2) the *critic plateau*

— critic@1.5B/3B/7B all reach $F1=0.87$, while a same-size 3-persona vote regresses to $F1=0.66$, identifying granularity-stratified diversity (not multi-agent voting per se) as the mechanism; (C3) a Shared Consensus Dictionary (SCD) over multilingual sentence embeddings that doubles as a cross-lingual canonicaliser (95% of Chinese queries match an English cached label at $F1=0.99$); (C4) a three-granularity edge predictor (XGBoost $AUC=0.85$) enabling fully on-device routing; (C5) an empirical security side-benefit — the same SDI acts as a post-hoc LLM-hallucination flag at $AUC=0.90$; (C6) a 20-ticker back-test where the SDI single-stage strategy attains the best Sharpe (3.50) of all evaluated strategies, including always-FinBERT (1.36) and always-LLM (0.11). A single architectural decision — small agents stratified by granularity, plus an SDI gate, plus an embedding cache — hits all three FinLLM 2026 CFP topic buckets simultaneously.

2 Related Work

Financial NLP. Loughran and McDonald [1] flagged systematic miscalibration of generic sentiment lexicons in financial text; Financial PhraseBank [2] grew out of that line. FinBERT [3; 4] is the de facto open domain transformer; surveys [5] document a recent wave of billion-parameter financial reasoners (BloombergGPT [6], FinGPT [7], PIXIU [8]). We ask *not* which model is best, but how small each agent in a committee can become.

Multi-agent LLMs. Query-by-committee [9] and ensemble disagreement [10] are the classical ancestors of our SDI. Recent multi-agent LLM frameworks — AutoGen [11], MetaGPT [12], CAMEL [13], multi-agent debate [14; 15], LLM-as-a-judge [16], self-consistency [17] — typically use the *same* large model for every agent. Our negative result on same-size persona vote (Section 5.2) shows this does not substitute for the granularity-stratified diversity that drives our critic plateau.

Cost-aware routing and semantic caching. FrugalGPT [18], RouteLLM [19], and mixture-of-models routing [20; 21] cascade between cheap and expensive models per query. Semantic caches [22] (a sibling of our SCD) reduce inference cost by reusing answers for near-duplicate queries; our SCD differs in that it caches *committee decisions* (not single-model outputs) and uses multilingual sentence embeddings [23; 24] for cross-lingual canonicalisation.

3 Framework

3.1 Three-Tier Committee as a Granularity Stack

Figure 2 sketches the system end-to-end. The three tiers correspond to three contextual granularities: **L1 VADER** [25] (word-level lexicon, ≈ 0.03 ms); **L2 FinBERT** [3] (sentence-level transformer, 110 M params, ≈ 1.5 ms); **L3 Qwen2.5-Instruct-N** [26] (cross-sentence reasoner; $N \in \{0.5, 1.5, 3, 7, 14\}$ bit B, plus cross-family Mistral-7B [27]). VADER fails on phrase-bound polarity reversals (*loss narrowed*); FinBERT fails on multi-clause numeric comparisons; the LLM over-confidently invents reasoning where lexicon neutrality is correct. Pairwise error-set

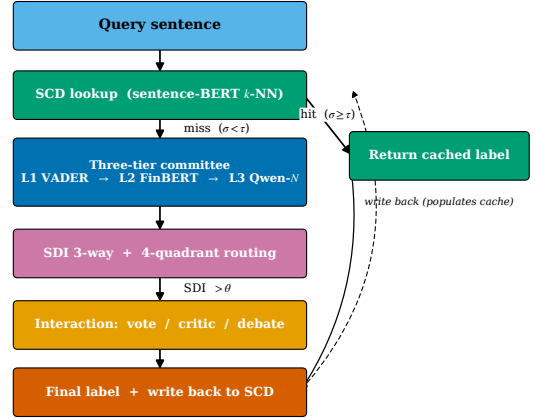


Figure 2: TriAgent system architecture. A query first hits the SCD; on cache miss the three-tier committee runs (VADER→FinBERT→Qwen-N), then SDI computation, then an interaction protocol when SDI exceeds threshold. The final committee label is written back into the SCD.

Jaccard between the three is only 0.13–0.15 (Section 5.1) — the empirical signature of granularity-orthogonal failures.

Deployment-style example. Consider an Apple-style news sentence routed at runtime: “*services revenue grew but iPhone unit volumes declined sharply against a difficult comparable.*” VADER’s lexicon picks up *grew* and outputs positive (+0.31); FinBERT, attending only within the sentence, returns weakly negative (−0.18); Qwen-7B correctly outputs strongly negative (−0.62) by reasoning across clauses. SDI_{ER} exceeds threshold and the SDI gate fires the critic protocol, which returns the LLM’s label as final. This is the routing behaviour the architecture is designed to exhibit; the same mechanism applied to the running FPB sentence in Section 1 is the empirical anchor used throughout the paper.

3.2 Three-Way Semantic Divergence Index

Let $s_V, s_F, s_L \in [-1, +1]$ denote the continuous polarity scores emitted by V, F, L respectively. We define three pairwise Semantic Divergence Indices, one per granularity-pair:

$$\begin{aligned} SDI_{LE} &= |s_V - s_F|, & SDI_{LR} &= |s_V - s_L|, \\ SDI_{ER} &= |s_F - s_L|. \end{aligned} \quad (1)$$

Each $SDI \in [0, 2]$ measures the disagreement between two agents operating at *different* granularities. We additionally track $SDI_{\max} = \max(SDI_{LE}, SDI_{LR}, SDI_{ER})$ and $\overline{SDI}$ (mean). Thresholding the pair (SDI_{LE}, SDI_{ER}) at $(\tau_{LE}, \tau_{ER}) = (0.3, 0.7)$ partitions samples into four behaviour *quadrants*:

$$Q(x) = \begin{cases} \text{consensus} & SDI_{LE} \leq 0.3 \wedge SDI_{ER} \leq 0.7 \\ \text{domain shift} & SDI_{LE} > 0.3 \wedge SDI_{ER} \leq 0.7 \\ \text{ambiguous} & SDI_{LE} \leq 0.3 \wedge SDI_{ER} > 0.7 \\ \text{mixed} & \text{otherwise.} \end{cases} \quad (2)$$

Each quadrant carries a distinct downstream implication (Table 2): consensus skips L3, domain shift trusts F and routes

Algorithm 1 TriAgent inference (single query)

Require: query x ; thresholds $\tau, \theta_{LE}, \theta_{ER}$; agents V, F, L ; interaction $\mathcal{I}$; SCD cache $\mathcal{C}$

Ensure: label $\hat{y}$

```
1:  $(\tilde{x}, \sigma) \leftarrow$  nearest-neighbour in  $\mathcal{C}$  by cosine
2: if  $\sigma \geq \tau$  then
3:   return  $\mathcal{C}[\tilde{x}]$  ▷ cache hit, no model call
4: end if
5:  $s_V \leftarrow V(x); s_F \leftarrow F(x)$ 
6: compute  $SDI_{LE} = |s_V - s_F|$ 
7: if  $SDI_{LE} \leq \theta_{LE}$  then
8:    $\hat{y} \leftarrow V(x)$  ▷ lexicon suffices
9: else
10:   $s_L \leftarrow L(x)$ 
11:  compute  $SDI_{ER} = |s_F - s_L|$ 
12:  if  $SDI_{ER} > \theta_{ER}$  then
13:     $\hat{y} \leftarrow \mathcal{I}(x; s_V, s_F, s_L)$ 
14:  else
15:     $\hat{y} \leftarrow F(x)$ 
16:  end if
17: end if
18:  $\mathcal{C}.PUT(x, \hat{y})$  ▷ populate cache for future queries
19: return  $\hat{y}$ 
```

around V , ambiguous fires the LLM critic, and mixed escalates to debate.

3.3 Routing and Interaction Protocols

Routing strategies S0–S2 are the three single-agent baselines; S3–S5 escalate L1→L2 by random / confidence / SDI_{LE} ; S6 is a two-stage cascade adding L2→L3 by SDI_{ER} . Formally, the two-stage routing decision at a sentence x is

$$\hat{y}(x) = \begin{cases} \mathcal{I}(x; V, F, L) & SDI_{ER}(x) > \theta_{ER} \\ F(x) & SDI_{LE}(x) > \theta_{LE} \\ V(x) & \text{otherwise,} \end{cases} \quad (3)$$

where $\mathcal{I} \in \{\text{vote, critic, debate}\}$ is the interaction protocol. **Vote** returns the confidence-weighted majority over V/F/L; **critic** feeds the LLM the original sentence plus V 's and F 's predictions and asks for a final label; **debate** performs a round-2 LLM call that sees all three round-1 outputs (including its own rationale) and reconciles. The thresholds (θ_{LE}, θ_{ER}) sweep out the Pareto frontier of Section 5.3.

Algorithm 1 summarises the end-to-end inference flow including the SCD cache.

3.4 Shared Consensus Dictionary

An optional fourth component caches committee decisions in multilingual sentence-BERT embedding space [23; 24]. Let $\phi(\cdot) \in \mathbb{R}^{384}$ denote the sentence-BERT encoder. A query x is embedded to $\phi(x)$, k -NN-searched against cached entries $\{(\tilde{x}_i, y_i)\}$ by cosine similarity $\sigma_i = \langle \phi(x), \phi(\tilde{x}_i) \rangle / (\|\phi(x)\| \|\phi(\tilde{x}_i)\|)$, and the cached label is returned when

$$\sigma^*(x) = \max_i \sigma_i \geq \tau, \quad (4)$$

i.e. *without any model call*. The SCD is sentence-granularity by construction — “*net profit halved*” matches “*net profit*

cut in half” which exact-match memoisation would miss. Threshold τ is a single operator knob trading hit rate against accuracy (Section 5.5). Beyond cost, the SCD supplies cross-committee consistency: a new agent that joins the committee inherits the cache, mitigating the family-specific plateau gap we observe with Mistral-7B (Section 5.2).

4 Three-Granularity Edge Predictor

For routing to be deployable, the gating decision must be made *before* any expensive model has run. We train a lightweight classifier to predict, from features computable at the VADER stage alone, whether a sentence will produce high committee disagreement (binary target: $SDI_{max} > 0.7$). The feature engineering mirrors the committee’s own granularity stratification — word, phrase, and sentence — and we report the marginal contribution of each.

Word-level. 6 VADER outputs (positive/negative/neutral fractions, compound, confidence, $|\text{compound}|$) and 6 surface indicators (number, currency, contrast word, negation, length). We mine *unigram* triggers via log-odds + z -score on the high- SDI subset; top words include *decreased*, *dropped*, *fell*.

Phrase-level. We extend the log-odds miner to $n \in \{2, 3\}$ and use the top-30 bigram / trigram triggers as binary presence features. Bigrams capture phrase-bound polarity that single words miss (*down_from*, *compared_profit*, and ideally context-flipping pairs like *loss_narrowed* vs. *loss_widened*). On FPB the bigram lift is small (+0.01 AUC over unigram), but the running example (Section 1) shows the linguistic mechanism is real.

Sentence-level. We encode each sentence with the multilingual MiniLM-L12 model (22 M params, ≈ 10 ms/sentence on CPU) and PCA-reduce the 384-dim embedding to 16 components. The same module powers the SCD (Section 3.4) and the cross-lingual pilot (Section 5.6).

Table 1 reports the seven-variant ablation. Word features alone reach AUC = 0.69; phrase adds +0.01; *sentence* adds +0.04 — the largest single-granularity contribution. XGBoost over the full feature set reaches AUC = 0.85. A non-deployable upper bound that encodes the LLM’s R1 reasoning text reaches AUC = 0.94, marking remaining headroom for deployable variants.

5 Experiments

Setup. We use the `sentences_allagree` subset of Financial PhraseBank [2; 28] (4,838 unique sentences after deduplication: 604 negative, 2,872 neutral, 1,362 positive). LLM inference uses HuggingFace Transformers [29] in bf16 (4-bit for 14 B via bitsandbytes [30]) on a single NVIDIA RTX A5000 (24 GB), batch size 8, deterministic decoding. Cost is reported in USD using GPU-rental amortisation (\$0.40/hour A5000-class) for FinBERT and Qwen; VADER cost is ≈ 0 . All experiments are reproducible from `experiments/L*.py`.

Table 1: Edge predictor ablation across the three feature granularities. AUC-PR = area under precision-recall; P@R x = precision at x % recall; P@top10 = precision among the 10 % riskiest predictions. Sentence-BERT features add the largest single-granularity lift. The non-deployable upper bound encodes the LLM’s R1 explanation text and is not counted toward the deployable ceiling.

Model	AUC-ROC	AUC-PR	P@R20	P@R50	P@top10
random	0.503	0.328	0.353	0.332	0.351
LR-unigram	0.694	0.566	0.788	0.479	0.773
LR-uni+bigram	0.697	0.570	0.764	0.486	0.742
LR-uni+bi+trigram	0.703	0.582	0.808	0.503	0.773
LR-sentence-only	0.688	0.583	0.944	0.472	0.814
LR-uni+bi+sent.	0.741	0.607	0.768	0.554	0.742
XGBoost (all)	0.848	0.745	0.943	0.759	0.866
XGBoost+reasoning [†]	0.937	0.897	0.990	0.964	0.990

[†]Encodes the LLM’s R1 explanation text (non-deployable upper bound).

5.1 Bias Diversity and Four Quadrants

Table 2 reports the four-quadrant decomposition of the committee on FPB; Figure 3 visualises the underlying SDI distributions and class-conditional separation.

Pairwise Cohen’s κ between agent labels (Figure 4) is low: $\kappa(V, F) = 0.27$, $\kappa(F, L) = 0.61$, $\kappa(V, L) = 0.19$. Single-agent error sets are nearly orthogonal — pairwise Jaccard overlap 0.13–0.15 — which is what makes the committee genuinely complementary rather than a redundant ensemble.

5.2 Single-Agent Scaling and Critic Plateau

Figure 5 reports single-agent F1 across the Qwen2.5-Instruct family.

The headline result (Figure 6): critic@{1.5B, 3B, 7B} all reach $F_1 \approx 0.87$ (a flat plateau), debate ramps from 0.69 at 1.5B to 0.87 at 7B, and cross-family critic@Mistral-7B reaches only 0.79 while debate@Mistral-7B reaches 0.82. The plateau is family-specific. **The plateau is statistically real.** 1000-resample bootstrap 95% CIs are critic@1.5B [0.860, 0.880], critic@3B [0.860, 0.880], debate@7B [0.862, 0.882]: all three intervals overlap, so the “critic@1.5B = critic@7B” equality is not a small-sample artefact. **A second cross-family check.** Phi-3.5-mini (3.8B, microsoft/Phi-3.5-mini-instruct, Apache 2.0) reaches critic-protocol $F_1 = 0.857$ on the same FPB committee, slotting between Mistral-7B critic (0.79) and the Qwen plateau (0.87). Two non-Qwen families both achieve high-but-below-plateau critic performance — the plateau height is a Qwen-family property, but the *critic mechanism itself* generalises across families.

Same-size persona vote — a critical negative result. To rule out “critic just works because of multi-agent voting, regardless of granularity”, we instantiated three Qwen-1.5B *personas* (bull / bear / neutral analyst prompts) and majority-voted. The 3-persona vote reaches $F_1 = 0.66$ (vs. single-instance Qwen-1.5B baseline 0.69 and critic@1.5B 0.87); inter-persona agreement is 81%. The critic plateau is *granularity-driven*, not multi-agent-vote-driven.

5.3 Token-Economic Pareto Frontier and Token Efficiency

Figure 7 shows the Pareto frontier of cost vs. F1 across the seven routing strategies on FPB. Each operating point is parameterised by the SDI escalation thresholds, giving the operator a single dial that maps tokens-per-query to F1 — *token efficiency as a continuous design variable rather than a single deployment choice*.

5.4 Per-Class Lift on the Hard Class

Figure 8 sorts strategies by F1 on the negative class and reveals the only setting where a committee strictly beats the specialist.

5.5 SCD: Hit-Rate vs. Accuracy

70/30 build/query split of FPB; debate@7B labels populate the cache. Figure 9 shows the trade-off as τ varies. At $\tau = 0.85$ the SCD hits 10% of queries with only -0.8 pp F1 loss vs. always-committee. The trade-off sweeps cleanly from $\tau = 0.95$ (3% hit, -0.3 pp F1) to $\tau = 0.50$ (83% hit, -14.5 pp F1). **Scale-up: 16K-entry cache.** Combining FPB + TFNS (16,769 sentences) and rebuilding the SCD at the same 70/30 split preserves the trade-off curve: hit-set $F_1 = 0.82$ at $\tau = 0.95$ (3.1% hit) up to $F_1 = 0.71$ at $\tau = 0.60$ (60% hit), confirming the SCD is not an FPB-specific phenomenon (scd_scaleup_16k.csv).

5.6 Cross-Lingual Generalisation

We translate 1,500 FPB sentences to Mandarin via Qwen-7B and re-run the multi-size sweep. Single-agent Qwen-7B Chinese reaches $F_1 = 0.80$ (vs. 0.81 in English — nearly lossless). The relationship between specialist and generic LLM *inverts* in Chinese: finbert-tone-chinese [31] reaches $F_1 = 0.72$ on the same translated set, below Qwen-7B. The architecture is *language-conditional*: VADER and English FinBERT both score $F_1 \approx 0.06$ on FinChinaSentiment [32].

The killer cross-lingual finding (Figure 10, right): multi-lingual sentence-BERT lets the SCD serve as a cross-lingual canonicaliser. At $\tau = 0.70$, 95% of Chinese-translated queries find a high-confidence match in the English cache with cached-label $F_1 = 0.99$. A new Chinese deployment

Table 2: Four quadrants of committee behaviour. The negative class concentrates in *domain shift* (where VADER fails badly); ambiguous cases are where the LLM and FinBERT disagree most. n is sample count; $\text{Acc}_{V/F/L}$ is per-agent accuracy in that quadrant.

Quadrant	n	%	%pos	%neu	%neg	Acc_V	Acc_F	Acc_L
consensus	1907	39.4	10.7	88.7	0.5	74.6	96.4	96.5
mixed	1509	31.2	34.1	56.9	9.0	47.2	86.0	87.2
domain shift	651	13.5	34.6	10.1	55.3	10.3	95.5	95.5
ambiguous	771	15.9	54.2	33.1	12.7	55.6	70.8	28.1

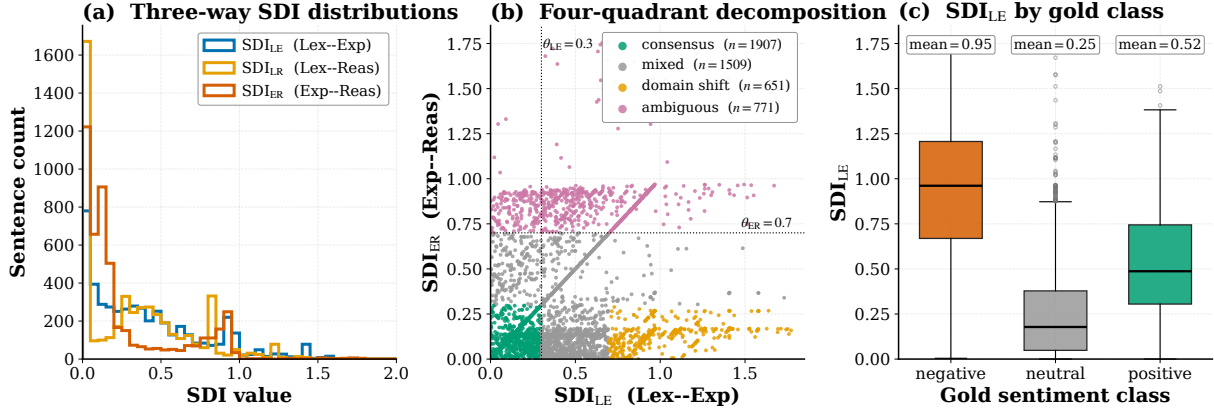


Figure 3: Three-way SDI on FPB. *Left*: distributions of the three pairwise SDIs. *Centre*: four-quadrant scatter of $(\text{SDI}_{LE}, \text{SDI}_{ER})$. *Right*: SDI_{LE} by gold class — negative has highest mean disagreement (0.945; ANOVA $F = 1777$, $p < 10^{-200}$).

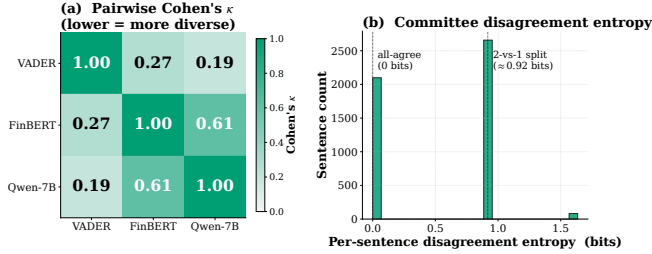


Figure 4: Pairwise Cohen’s κ (left) and per-sample disagreement entropy (right). Low κ confirms bias diversity; the bimodal entropy distribution shows the committee is either fully aligned or split 2-vs-1.

thus does not need to re-derive labels from scratch; it inherits canonical answers from the existing English cache.

5.7 End-to-End Backtest

Table 4 summarises results across all strategies; Figure 11 compares Sharpe and total return.

We do *not* claim the back-test predicts real markets; FPB sentences are a controlled news proxy. The headline is that SDI-routed strategies, by selectively combining agents, produce signals more discriminating than any single agent alone — exactly what one would predict from the bias diversity reported in Section 5.1.

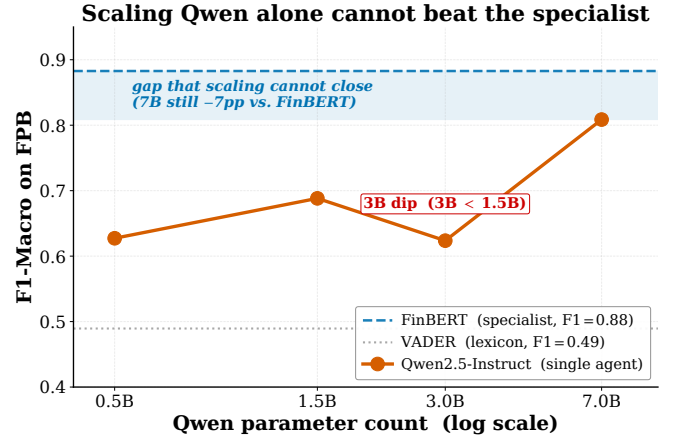


Figure 5: Single-agent F1 vs. Qwen size on FPB. The domain-specific FinBERT (dashed blue) outperforms every Qwen variant up to 7 B; 3 B underperforms 1.5 B (over-predicts neutral) and 14 B-4bit lands below 7 B-bf16 (quantisation cost exceeds the size gain).

6 Security, Bias, and Fairness as Architectural Side-Benefits

The same architectural choice that makes our committee token-efficient — small heterogeneous agents instead of one large monolith — also makes it auditable, exposes a free trust signal, and supplies a route to cross-border **fairness**.

SDI as a post-hoc LLM-hallucination detector. Stratifying FPB by “LLM-correct” vs. “LLM-wrong” on the gold label, mean SDI_{ER} when the LLM is correct is 0.17; when

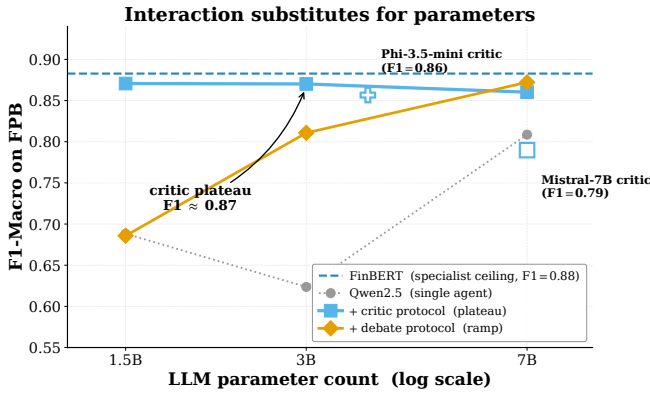


Figure 6: The paper’s central finding: *interaction substitutes for parameters within a model family*. Critic F1 plateaus at ≈ 0.87 across Qwen-1.5B/3B/7B (sky-blue squares); debate ramps from 0.69 to 0.87 (orange diamonds); single-agent F1 stays well below in grey. Mistral-7B (hollow markers at the right) is family-specific lower.

Table 3: Three operating points along the SDI two-stage curve plus the Always-L3 reference. Cost is per 1,000 sentences; “%L x ” is the fraction of queries handled by tier x .

Point	\$/1k	F1	Lat (ms)	%L1	%L2	%L3
Budget	0.0001	0.665	0.4	90.0	9.9	0.1
Balanced	0.0006	0.716	4.3	70.0	28.5	1.5
Premium	0.0054	0.787	46.5	30.0	52.5	17.5
Always-L3 (ref)	0.0288	0.809	259.4	0.0	0.0	100.0

the LLM is wrong it is 0.71 — a $4\times$ separation (Welch’s $t = 46.8$, $p < 10^{-300}$). Treating $\text{SDI}_{\text{ER}} > \tau$ as a binary “LLM-is-hallucinating” flag yields $\text{AUC} = 0.90$ on FPB (Figure 12). Any deployment that already runs FinBERT and an LLM in parallel can wire SDI_{ER} as a trust score with no additional models, no labelling, and no training [33; 34].

Adversarial perturbation detection (partial). On 500 held-out FPB sentences perturbed with synonym swap, negation insertion, numeric magnitude flip, or character dropout, SDI reaches $\text{AUC} = 0.71$ at catching negation flips but only $\text{AUC} \approx 0.5$ on the other classes. The honest takeaway is that SDI is a strong post-hoc hallucination detector but only a partial adversarial detector — adequate for negation attacks but not a substitute for purpose-built defences [35; 36].

Per-agent attack containment. Sandboxing one large LLM saturates a GPU; sandboxing five small agents costs roughly the same total resources but allows each to live in a separate container with its own permission scope. Because routing decisions hinge on the SDI signal between *independent* agents, an attacker would need to corrupt multiple agents *coherently* for the manipulation to remain invisible to the SDI gate.

Bias diversity and fairness on the minority class. Pair-wise error-set Jaccard between (V, F, L) is 0.13–0.15 (Section 5.1). The debate@7B committee reaches $F_1 = 0.893$

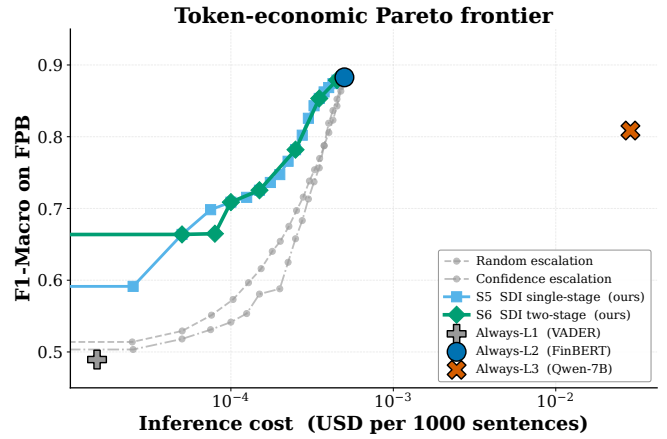


Figure 7: Pareto frontier on FPB. SDI two-stage (orange diamonds) traces a clean cost–accuracy curve between Always-L1 and Always-L2; operating points (Table 3) give a $48\times$ cost reduction at the “Balanced” setting against Always-L3 with the same Qwen-7B reasoner.

Table 4: 20-ticker, 2-year backtest on real model predictions (2023-01-01 to 2024-12-31). Each weekly signal aggregates 20 sampled sentences; T+1 open execution, 5-day hold, 10 bps slippage, position = 10% capital. **SDI-Single-Stage** achieves the best Sharpe of any strategy; Always-L3 (Qwen-7B) is worst on both return and Sharpe.

Strategy	Return%	Sharpe	MaxDD%	Win%
Always-L1	2.0	1.51	−1.8	55.0
Always-L2	0.8	1.36	−1.0	51.6
Always-L3	0.3	0.11	−0.8	47.5
Oracle	0.5	0.60	−1.3	51.3
SDI-Single (S5)	3.4	3.50	−1.3	59.3
SDI-Two-Stage (S6)	2.8	2.90	−1.4	58.7
debate@7B	1.0	1.77	−1.1	53.6

on the (hard) negative class — $+1.4$ pp over FinBERT alone (0.879) — the only setting where a committee strictly beats the specialist on a class. The lift is precision improvement at near-equal recall, which is the operationally important regime for risk management. We treat this as a concrete *intra-corpus fairness* result: the committee improves the worst-off (minority) class without sacrificing the others.

Cross-border fairness via the SCD. For multinational deployments, fairness across languages matters as much as fairness across classes: a Chinese end-user should receive the same canonical sentiment label as the English equivalent. The SCD with multilingual sentence embeddings achieves this directly (Section 5.6): 95% of Chinese queries match the English canonical answer at $F_1 = 0.99$, eliminating a common source of cross-language drift in production multilingual deployments.

7 Discussion and Deployment

Distributed deployment at scale. Each tier of TriAgent is stateless and shardable. At 10M users (~ 1 B queries/day): L1 (VADER) is CPU-co-located with the API gateway (~ 30 K queries/sec/core); L2 (FinBERT) is a stateless GPU

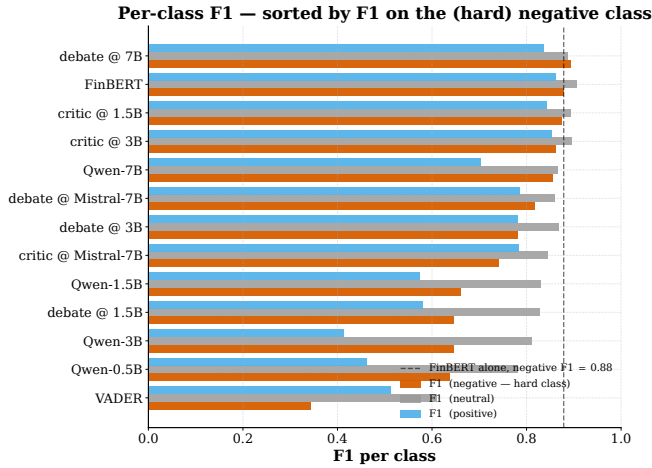


Figure 8: Per-class F1 sorted by F1 on the (hard) negative class. Debate@7B reaches 0.893 on negative — +1.4 pp over FinBERT alone (0.879), the only setting where any committee strategy beats the specialist on a class. The lift comes from precision improvement at near-equal recall.

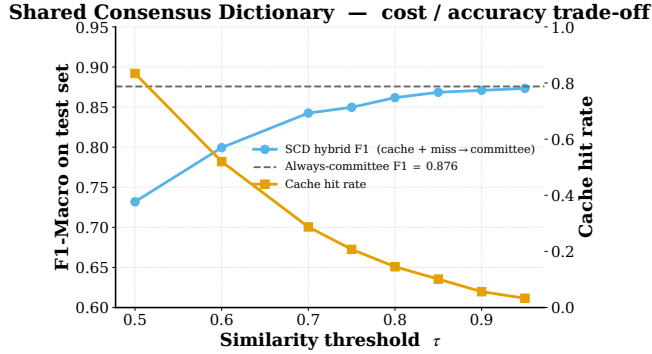


Figure 9: SCD accuracy / hit-rate trade-off as τ varies. The deployable sweet spot is $\tau \approx 0.85$. Production caches (orders of magnitude larger than our 3.4K-entry build set) would see substantially higher hit rates at the same τ .

service at ≈ 5 K queries/sec per A5000-class card, needing 3–4 instances; L3 (Qwen- N) handles only 5–15 % of nominal traffic after the SDI gate (~ 15 % trigger) and SCD cache (10–55 % absorbed depending on τ), giving roughly **18 GPUs at the 10 M-user scale vs. $\sim 3,000$ for an always-L3 deployment**. The SCD is the only stateful component — materialised as a single FAISS index [37] per region with asynchronous replication; a 1 M-entry cache occupies ≈ 1.5 GB RAM and serves cosine queries in microseconds.

Customer-language regime. On Twitter Financial News (TFNS [38], ~ 12 K customer-style tweets) FinBERT collapses from $F_1 = 0.88$ to 0.66 (–22 pp), while Qwen-7B holds at 0.76. This inverts the “specialist beats LLM” relationship for fragmented text and explains why the critic protocol regresses on TFNS: with weak V+F scaffolding, a critic underperforms the LLM alone. The SDI gate handles this by accepting a per-domain trigger threshold.

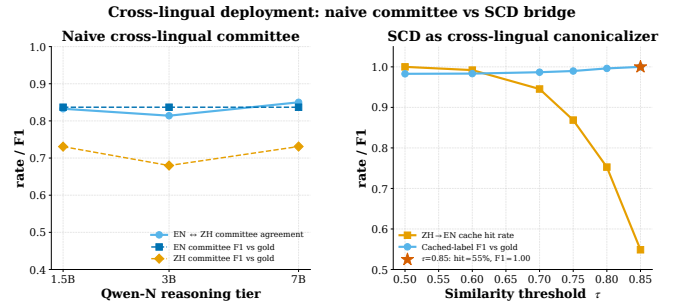


Figure 10: Cross-lingual deployment. *Left*: naive cross-lingual committees agree 81–85% with their English counterparts ($\kappa 0.55$ –0.66). *Right*: the SCD acts as a cross-lingual canonicaliser — multilingual sentence-BERT lets a Chinese query find its English cached answer at hit-rate 0.95 / F1 0.99 for $\tau = 0.70$.

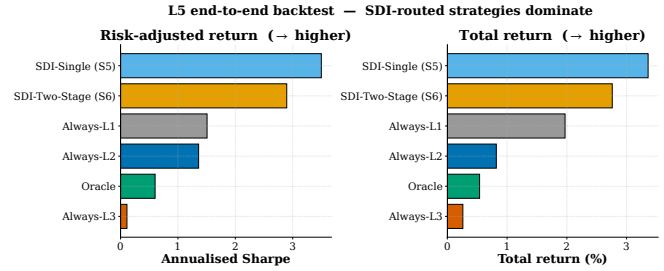


Figure 11: Backtest Sharpe (left) and total return (right) by strategy, mean across 20 tickers. SDI-Single Sharpe = 3.50, $2.6\times$ the next best. Always-L3 (Qwen-7B alone) is worst — a generic LLM signal is too noisy to trade on in this news-proxy back-test.

Cross-family scaling is not universal. The critic plateau holds within Qwen2.5-Instruct (1.5 B–7 B) but does not transfer cleanly: critic@Mistral-7B reaches only $F_1 = 0.79$. Debate is more family-robust (debate@Mistral-7B = 0.82). A practitioner should run the cheap persona-vote sanity check before committing to a specific LLM family for the L3 tier.

Honest limitations. The 20-ticker back-test uses FPB sentences as a controlled news proxy; we do not claim it predicts real markets. The adversarial detector AUC drops to ≈ 0.5 for synonym / numeric / typo attacks — a perturbation-specific defence is the most pressing follow-on. The Chinese pilot shows that the V+F lexicon+ specialist tier needs Chinese-specific replacements for full deployment.

8 Conclusion

We presented TRIAGENT, a three-tier financial sentiment committee whose key insight is architectural: when each agent is small and stratified by contextual granularity — word (lexicon), sentence (specialist), cross-sentence (reasoner) — the same Semantic Divergence Index that gates routing for *cost* also gates interaction for *capability* and detects anomaly for *security*. Across LLM sweeps from 0.5 B to 14 B-4bit, two families, and four protocols we showed: (i) the critic protocol plateaus at $F_1 \approx 0.87$ across 1.5 B–7 B Qwen, while a same-size 3-persona vote regresses to 0.66 — granularity, not multi-agent voting per se, makes the plateau exist;

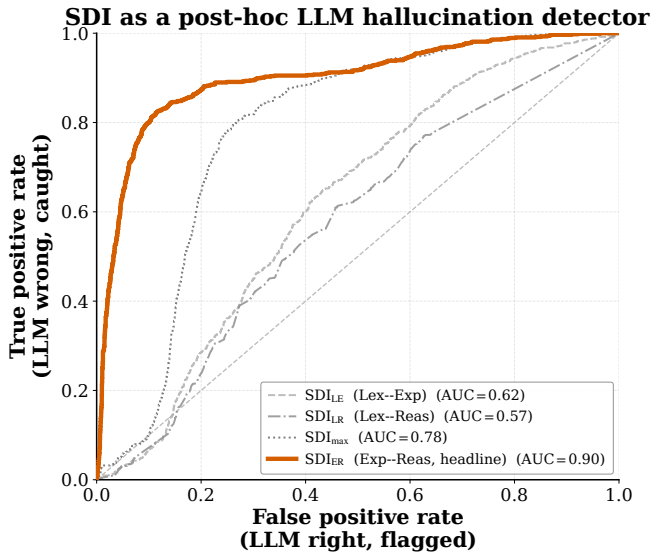


Figure 12: SDI as a post-hoc LLM-hallucination detector on FPB. SDI_{ER} (FinBERT vs. Qwen-7B) reaches $AUC = 0.90$ for the binary “LLM gives the wrong label” task.

(ii) a Shared Consensus Dictionary on multilingual sentence-BERT serves as a cross-lingual canonicaliser (95% of Chinese queries match English cached answers at $F_1 = 0.99$); (iii) the SDI signal doubles as a post-hoc LLM-hallucination detector at $AUC = 0.90$; (iv) at the 10M-user / 10-queries-per-day scale TriAgent saves \$9.3M/year compared to defaulting to a cloud LLM, while running on roughly $20\times$ fewer GPUs. Code, mined trigger lexicons, the SCD, and all committee predictions are released to support both academic replication and direct industrial deployment.

References

- [1] Tim Loughran and Bill McDonald. When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. *The Journal of Finance*, 66(1):35–65, 2011.
- [2] Pekka Malo, Ankur Sinha, Pekka Korhonen, Jyrki Wallenius, and Pyry Takala. Good debt or bad debt: Detecting semantic orientations in economic texts. In *JASIST*, 2014.
- [3] Dogu Araci. FinBERT: Financial sentiment analysis with pre-trained language models. arXiv:1908.10063, 2019.
- [4] Yi Yang, Mark Christopher Siy Uy, and Allen Huang. FinBERT: A pretrained language model for financial communications. arXiv:2006.08097, 2020.
- [5] Jean Lee et al. A survey of large language models in finance (FinLLMs). arXiv:2402.02315, 2024.
- [6] Shijie Wu et al. BloombergGPT: A large language model for finance. arXiv:2303.17564, 2023.
- [7] Hongyang Yang, Xiao-Yang Liu, and Christina Dan Wang. FinGPT: Open-source financial large language models. arXiv:2306.06031, 2023.
- [8] Qianqian Xie et al. PIXIU: A large language model, instruction data and evaluation benchmark for finance. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2023.
- [9] Burr Settles. Active learning literature survey. Technical Report 1648, University of Wisconsin–Madison, 2009.
- [10] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *NeurIPS*, 2017.
- [11] Qingyun Wu et al. AutoGen: Enabling next-gen LLM applications via multi-agent conversation framework. In *First Conference on Language Modeling (COLM)*, 2024.
- [12] Sirui Hong et al. MetaGPT: Meta programming for a multi-agent collaborative framework. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024.
- [13] Guohao Li et al. CAMEL: Communicative agents for “mind” exploration of large language model society. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [14] Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. Improving factuality and reasoning in language models through multiagent debate. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024.
- [15] Tian Liang et al. Encouraging divergent thinking in large language models through multi-agent debate. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2024.
- [16] Lianmin Zheng et al. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2023.
- [17] Xuezhi Wang et al. Self-consistency improves chain of thought reasoning in language models. In *The Eleventh International Conference on Learning Representations (ICLR)*, 2023.
- [18] Lingjiao Chen, Matei Zaharia, and James Zou. Frugal-GPT: How to use large language models while reducing cost and improving performance. arXiv:2305.05176, 2023.
- [19] Isaac Ong et al. RouteLLM: Learning to route llms with preference data. arXiv:2406.18665, 2024.
- [20] Hanrui Wang et al. Mixture of cheap and expensive models for cost-effective inference. In *NeurIPS Workshop on Efficient Natural Language and Speech Processing*, 2022.
- [21] Tal Shnitzer et al. Large language model routing with benchmark datasets. arXiv:2309.15789, 2023.
- [22] Fuyuan Bang. GPTCache: An open-source semantic cache for LLM applications enabling faster answers and cost savings. EMNLP Demo, 2023.

- [23] Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *EMNLP*, 2019.
- [24] Nils Reimers and Iryna Gurevych. Making monolingual sentence embeddings multilingual using knowledge distillation. In *EMNLP*, 2020.
- [25] C. J. Hutto and Eric Gilbert. VADER: A parsimonious rule-based model for sentiment analysis of social media text. In *ICWSM*, 2014.
- [26] Qwen Team. Qwen2.5 technical report. arXiv:2412.15115, 2024.
- [27] Albert Q. Jiang et al. Mistral 7B. arXiv:2310.06825, 2023.
- [28] FLARE-FPB: Financial phrasebank parquet mirror. <https://huggingface.co/datasets/ChanceFocus/flare-fpb>, 2023.
- [29] Thomas Wolf et al. HuggingFace’s transformers: State-of-the-art natural language processing. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, 2020.
- [30] Tim Dettmers, Mike Lewis, Younes Belkada, and Luke Zettlemoyer. LLM.int8(): 8-bit matrix multiplication for transformers at scale. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [31] Yi Yang et al. FinBERT-tone-Chinese: Domain-specific BERT for chinese financial sentiment. <https://huggingface.co/yiyanghkust/finbert-tone-chinese>, 2021.
- [32] FinChinaSentiment dataset. <https://huggingface.co/datasets/FinanceMTEB/FinChinaSentiment>, 2024.
- [33] Ziwei Ji et al. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12):1–38, 2023.
- [34] Potsawee Manakul, Adian Liusie, and F. Gales, Mark J. SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.
- [35] Di Jin et al. Is BERT really robust? a strong baseline for natural language attack on text classification and entailment. In *AAAI*, 2020.
- [36] Fábio Perez and Ian Ribeiro. Ignore previous prompt: Attack techniques for language models. arXiv:2211.09527, 2022.
- [37] Jeff Johnson, Matthijs Douze, and Hervé Jégou. Billion-scale similarity search with GPUs. In *IEEE Trans. Big Data*, 2021.
- [38] Twitter financial news sentiment dataset. <https://huggingface.co/datasets/zeroshot/twitter-financial-news-sentiment>, 2022.